

Example: Table 3-2 Truth table of Function F

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

In sum of minterms the function is expressed as: $F(x,y,z) = \Sigma(1,3,4,6)$

x\yz	00	01	11	10
0		1	1	
1	1			1

$$\text{SOP: } x'z + xz' = F$$

x\yz	00	01	11	10
0	0			0
1		0	0	

$$F': xz + x'z'$$

$$(F')' = F = (xz + x'z')' = (xz)' \cdot (x'z')' \text{ (Theorem 5(a))}$$

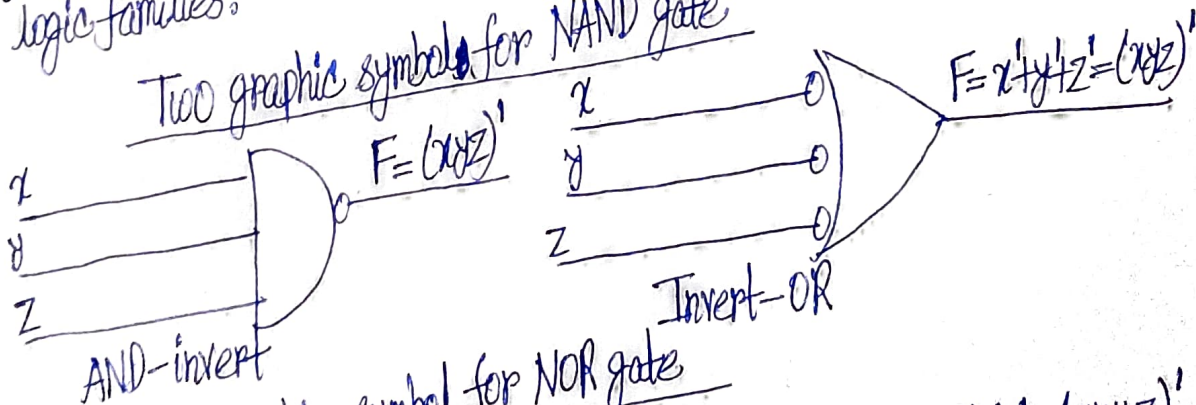
$$= (x' + z') \cdot (x + z) \text{ (Theorem 5(b))}$$

$$= (x' + z') \cdot (x + z) \text{ (Theorem 3)}$$

3.6 NAND and NOR Implementation

Digital circuits are more frequently constructed with NAND or NOR gates than with AND and OR gates. NAND and NOR gates are easier to fabricate with electronic components and are the basic gates used in all IC digital logic families.

Two graphic symbols for NAND gate



Two graphic symbol for NOR gate

